PDF Summarization Agent with Llama 3

EMEA AI Cohort 1 – Hands-on Assignment 11

1. Project Overview

The PDF Summarization Agent is an AI-powered application that extracts text from PDF documents and generates concise, meaningful summaries using the Llama 3 large language model.

The application uses PyMuPDF for PDF text extraction, Ollama to run Llama 3, FastAPI to provide a REST API, and Gradio to create a simple web interface for uploading documents and viewing summaries.

2. Objectives

- Upload and validate PDF documents.
- Extract readable text from PDF pages.
- Split long documents into manageable text chunks.
- Generate summaries using Llama 3.
- Provide a FastAPI endpoint for PDF summarization.
- Display results through an interactive web interface.
- Handle invalid files and processing errors gracefully.

3. Technologies Used

Technology	Purpose
Python	Application development
Google Colab	Development environment
PyMuPDF	PDF text extraction
Ollama	Local model serving
Llama 3	AI text summarization
FastAPI	Backend API
Uvicorn	API server
Gradio	Web-based user interface

Requests	HTTP communication

4. System Workflow

1. The user uploads a PDF document through the Gradio interface.
2. The document is sent to the FastAPI /summarize endpoint.
3. The application validates the uploaded file.
4. PyMuPDF extracts readable text from the PDF.
5. The extracted text is divided into chunks when necessary.
6. Llama 3 generates summaries for the text.
7. The partial summaries are combined into a final summary.
8. The result is returned by FastAPI and displayed in the Gradio interface.

5. Main Features

PDF Upload and Validation

Accepts PDF files and checks file type, size, page count, and readable text.

Text Extraction

Extracts text from PDF pages using PyMuPDF.

Text Chunking

Splits longer documents into smaller sections to make processing more manageable.

AI-Powered Summarization

Uses Llama 3 through Ollama to generate structured summaries containing important information.

FastAPI Integration

Provides an API for submitting PDF files and receiving summary results.

Interactive User Interface

Allows users to upload documents and view summaries through a browser-based interface built with Gradio.

Error Handling

Provides useful error messages for invalid PDFs, unreadable documents, connection problems, and summarization failures.

6. API Endpoints

Method	Endpoint	Description
GET	/	Checks API status
POST	/summarize	Accepts a PDF and returns its summary

The interactive API documentation is available at /docs while the FastAPI server is running.

7. Installation and Execution

The project is developed in Google Colab.

1. Install the required Python packages.
2. Install and start Ollama.
3. Download the Llama 3 model.
4. Define the PDF extraction, validation, and chunking functions.
5. Implement the summarization function.
6. Create the FastAPI application and its endpoints.
7. Start the Uvicorn server.
8. Test the API using a PDF document.
9. Launch the Gradio interface.
10. Upload a PDF and verify the generated summary.

8. Testing

The following tests should be performed:

- Upload a valid PDF and verify that a summary is generated.
- Confirm that the API returns HTTP 200 for a successful request.
- Upload a non-PDF file and verify that it is rejected.
- Test a PDF with insufficient readable text.
- Check how the application responds when Ollama is unavailable.
- Compare the generated summary with the original document to identify incorrect or missing details.

9. Limitations

- Scanned PDFs may require OCR to extract text from images.

- Summary generation may take longer for large documents.
- AI-generated summaries may occasionally omit or misinterpret information.
- Google Colab sessions and Gradio sharing links are temporary.
- The application requires the Ollama service and Llama 3 model to be available.

10. Future Enhancements

- Add OCR support for scanned PDFs.
- Allow users to select summary length.
- Support multiple languages.
- Add an option to download summaries as PDF or text files.
- Improve processing of very large documents.
- Deploy the application to a persistent hosting environment.

11. Conclusion

The PDF Summarization Agent demonstrates how large language models can be integrated with document-processing tools and web APIs to simplify the understanding of lengthy PDF documents.

By combining PyMuPDF, Ollama, Llama 3, FastAPI, and Gradio, the project provides an end-to-end workflow for uploading documents, extracting text, generating summaries, and presenting the results through a user-friendly interface.